

Exercises for Chapter 3

Connor Mooneyhan

July 9, 2025

Exercise 1. Prove that convergence of $\{s_n\}$ implies convergence of $\{|s_n|\}$. Is the converse true?

Proof. If $\{s_n\} \rightarrow s$, then let $\varepsilon > 0$ and $N \in \mathbb{Z}^+$ such that $n > N$ implies $|s_n - s| < \varepsilon$. Then

$$\left| |s_n| - |s| \right| \leq |s_n - s| < \varepsilon$$

by the reverse triangle inequality.

The converse hypothesis is contradicted by the alternating series $\{(-1)^{n+1}\}$, where the absolute value of the sequence clearly converges to 1, whereas the sequence itself diverges. $\square$

Exercise 2. Calculate $\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n)$.